

Field Overview: Disciplinary Structure and Growth Dynamics

The Active Inference literature has undergone a phase transition. What originated in the early 2000s—building on predictive coding and Bayesian brain foundations from the late 1990s—as a niche within theoretical neuroscience has expanded rapidly into a multi-disciplinary research program spanning three primary domains and eight tracked categories. The corpus start of 2005 was chosen to capture Energy-Based Model and variational Bayesian antecedents dayan1995helmholtz, lecueun2006tutorial that preceded the formal introduction of the Free Energy Principle in 2006 friston2006free and its subsequent full elaboration friston2010free. Our corpus, extracted from arXiv, Semantic Scholar, and OpenAlex and deduplicated to $N = 849$ papers (2005–2026), captures the breadth, tempo, and internal architecture of this expansion (Figure 1).

Active Inference Literature by Subfield (N=847)

